Arista MAR-ACC-36 Optimal Transport Failure Resolution SOP

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8499

Description

The Access Role on the MAR-ACC-36 node within the Marathahalli Core Site on Arista Equipment is susceptible to Optical Transport Failures due to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This is primarily attributed to the inherent characteristics of the Optical Transport Network (OTN) protocol. The OTN is a high-speed, point-to-point optical transmission technology that relies on the reliable exchange of signals between adjacent nodes. However, the physical nature of the OTN, involving the transmission of light signals through fiber-optic cables, renders it vulnerable to environmental factors such as temperature fluctuations, mechanical stress, and physical damage. Furthermore, the use of SFP/XFP modules, which act as interfaces between the OTN and the underlying physical transmission medium, introduces additional failure points. The degradation of these modules, resulting from prolonged exposure to environmental stressors or manufacturing defects, can lead to reduced signal quality and ultimately, total failure. As a result, the Access Role on the MAR-ACC-36 node is exposed to the risk of Optical Transport Failures, necessitating the implementation of a comprehensive SOP to diagnose and rectify such failures.

Pre-requisite Checks

- check: Verify the site location and node ID to ensure accuracy
- check: Confirm the equipment vendor and model to ensure compatibility
- check: Check the redundant link status to ensure no single point of failure

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: show interfaces status, Expected Output: Interface Status: up/down/admin down, Description: This command displays the current status of all interfaces on the device, allowing the operator to identify any potential issues related to the Optical Transport Network.

Command: show interfaces statistics, Expected Output: Interface Statistics: packets sent/received errors, Description: This command provides detailed statistics on interface activity, enabling the
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operator to isolate any anomalies or degradation in the OTN signal.

Command: show spanning-tree root, Expected Output: Root Bridge ID: [bridge ID], Description: This command displays the current root bridge ID, allowing the operator to verify that the OTN is functioning correctly and that there are no anomalies in the spanning tree topology.
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Resolution Steps

Step 1: Step: Swap the SFP/XFP module, Command: replace interfaces 1/1/1 with 1/1/2, Description: This step involves replacing the suspect SFP/XFP module with a new one, which should resolve the issue if the problem was caused by a faulty module.

Step 2: Step: Verify the fiber connections, Command: show interfaces fiber, Description: This step involves visually inspecting the fiber connections to ensure that they are secure and not damaged, as a loose or damaged fiber can cause signal degradation and failure.

Step 3: Step: Reset the OTN signal, Command: clear interfaces otu, Description: This step involves resetting the OTN signal to its default state, which should resolve any issues related to signal degradation or power loss.

Post-Check Verification

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Command: show interfaces status, Expected Output: Interface Status: up, Description: This command is used to verify that the interface is functioning correctly and that the issue has been resolved.

Command: show interfaces statistics, Expected Output: Interface Statistics: packets sent/received errors = 0, Description: This command is used to verify that the interface statistics are within normal ranges and that there are no errors.

Command: show spanning-tree root, Expected Output: Root Bridge ID: [bridge ID], Description: This command is used to verify that the OTN is functioning correctly and that there are no anomalies in the spanning tree topology.
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Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace interfaces 1/1/1 with 1/1/1 original, Description: If the replacement SFP/XFP module does not resolve the issue, this step involves restoring the original module to its previous state.

- Step: Verify the fiber connections, Command: show interfaces fiber, Description: This step involves visually inspecting the fiber connections to ensure that they are secure and not damaged.

- Step: Reset the OTN signal, Command: clear interfaces otu, Description: This step involves resetting the OTN signal to its default state.